

Project Proposal Report

Date	06 July 2026
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Team

Project Proposal (Proposed Solution) Report

Project Overview

Smart Lender transforms the loan approval system using a high-accuracy classification pipeline. It removes manual credit assessment delays by utilizing machine learning models trained on structural applicant features. Key aspects include real-time single and batch prediction, robust data imputation, and instant credit risk flagging.

Objective: Deploy an automated web-based evaluator resolving processing delays.

Scope: Implement offline data prep, SMOTE class balancing, scaler encoding, XGBoost scoring, and a light-mode Flask UI.

Resource Requirements & Allocation

Resource Type	Description	Specification/Allocation
Hardware	Computing Power	T4 GPU / 8 Cores CPU for model training
Hardware	Memory (RAM)	8 GB RAM minimum configuration
Hardware	Storage Space	1 TB SSD storage for dataset, logs, and pickles
Software	Web Framework	Python 3.13, Flask Framework
Software	Libraries	scikit-learn, xgboost, imbalanced-learn, pandas, numpy
Software	IDE	Jupyter Notebook, VS Code, PyCharm
Data	Data Source	Kaggle loan prediction dataset (614 rows, CSV format)